AETHERIS TECHNICAL PUBLICATION • SPEC-2026-AEOS

AETHERIS AEOS

Autonomous Enterprise Operating System

Complete Architectural Specification, Multi-Agent Swarm Dynamics, Cryptographic Ledger Verification, Statutory Multi-Jurisdictional Regulatory Compliance (EAR99, EU CBAM, German LkSG), and Enterprise Operational Implementation Manual.

DOCUMENT REF	PRODUCTION ENGINE	AUTHORITATIVE PORT	SECURITY MATRIX
AEOS-2.6.4-SPEC-2026	**FastAPI 0.115+ (Python)**	**Backend :3001 // Vite :5173**	**BFT + Ed25519 Merkle**

Executive Manual Contents

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01. Executive Summary & Core Paradigm

AETHERIS AEOS (Autonomous Enterprise Operating System) is a mission-critical, AI-orchestrated autonomous enterprise governance platform. In contemporary enterprise supply chains, organizations lose billions of dollars annually due to bureaucratic manual handoffs across disconnected ERP silos (SAP S/4HANA, Oracle NetSuite, Microsoft Dynamics 365). Procurement delays precipitate massive contractual Service Level Agreement (SLA) late penalties, illicit export control exposure, and unexpected carbon border liabilities.

AETHERIS AEOS transforms enterprise operating software from a passive relational ledger into an active, deterministic, multi-agent swarm. When supply chain shocks or inventory shortages emerge, AEOS detects the deficit within milliseconds, conducts parallel global supplier arbitrage, audits international export and carbon mandates with explicit statutory citations, enforces cryptographic separation of duties, and produces a mathematically provable Merkle audit trail.

CORE PARADIGM SHIFT

The Autonomous Triad: (1) Deterministic mathematical decisioning over probabilistic hallucination; (2) Strict four-eyes human governance over high-value capital allocation ($5,000 threshold); (3) Cryptographic non-repudiation using RFC 8785 canonical JSON, SHA-256 chaining, and Ed25519 digital signatures.

02. End-to-End System Architecture

AETHERIS AEOS is structured as a decoupled, high-throughput client-server topology. The backend is powered by an asynchronous **FastAPI (Python 3.11+)** engine operating strictly on port **3001**. The frontend is a reactive Single Page Application built on **React 19, TypeScript, and Vite** on port **5173**.

```
+-------------------------------------------------------------------------------------------------+ |
AETHERIS CLIENT APPLICATION (PORT 5173) | | React 19 + TypeScript 5.8 + Vite + Tailwind CSS + Lucide Icons +
Google GenAI SDK | | [Problem Explorer] [Swarm Visualizer] [Mission Control] [Compliance Radar] [Operations
Desk] | +-------------------------------------------------------------------------------------------------+ |
Vite Proxy: /api/* -> :3001 v
+-------------------------------------------------------------------------------------------------+ |
FASTAPI AUTHORITATIVE BACKEND (PORT 3001) | | [Auth & RBAC /api/auth] [Swarm Orchestrator] [Operations Engine]
[Compliance Sentinel] | |
------------------------------------------------------------------------------------------------- | |
CRYPTOGRAPHIC ENGINE: RFC 8785 Canonical JSON | SHA-256 Chaining | Ed25519 Merkle Tree Sealing | |
------------------------------------------------------------------------------------------------- | | PERSISTENCE
LAYER: SQLAlchemy ORM (14 Relational Models) -> SQLite 3 / PostgreSQL 16 |
+-------------------------------------------------------------------------------------------------+
```

LEGACY SERVER RETIREMENT NOTICE

Architectural Retirement: Earlier iterations included an experimental Node/Express server in server/. This server is formally retired. The authoritative enterprise backend is strictly **FastAPI on port 3001**.

03. Full Technology Stack & Dependencies

Tier	Component	Version	Operational Responsibility
Frontend	React Core	19.0.0	Concurrent rendering, fine-grained state management, transitions
Frontend	TypeScript	5.8.2	Strict type assertions across API contracts and agent states
Frontend	Vite	6.2.0	Fast HMR dev server, reverse-proxy /api routing to port 3001
Frontend	Lucide React	^1.16.0	Enterprise dashboard iconography across cockpits and radars
Frontend	@google/genai	^0.1.1	Direct client interaction with Gemini 2.5 Flash and Pro models
Backend	FastAPI	0.115+	High-speed async REST endpoints with native OpenAPI validation
Backend	Uvicorn	0.30+	Lightning-fast ASGI production web server running on port 3001
Backend	SQLAlchemy	2.0+	Type-safe Object Relational Mapping governing 14 database models
Backend	Pydantic	2.0+	High-throughput schema serialization and validation
Security	cryptography	42.0+	Ed25519 asymmetric signing & SHA-256 Merkle root computation
Database	SQLite / Postgres	3.x / 16	ACID relational transaction store and immutable audit ledger

04. Frontend Architecture & Views Catalog

The AETHERIS AEOS user interface is orchestrated by App.tsx as a high-density, real-time command cockpit. The platform houses six specialized operational views designed to provide zero-latency situational awareness:

View Route	Module Title	In-Depth Functional Capabilities
problem	**Problem Statement Explorer**	Quantifies systemic fragility in conventional enterprise resource planning. Models operational breakdown vectors: legacy ERP batch processing latencies (12-36 hrs), manual procurement friction, lack of carbon accounting, and contractual delivery penalty exposure. Features real-time ROI simulation sliders.
swarm	**Swarm Visualizer**	Interactive dynamic graph depicting the five active autonomous agent nodes. Displays live inter-agent message propagation, current sub-task dispatch queues, internal memory cache utilization, and real-time voting rounds over the communication bus.
mission	**Mission Control Cockpit**	The operational nervous system of AEOS. Displays real-time run IDs, progressive execution timelines, live system logs, agent chain-of-thought streams, BFT consensus voting matrices, and live Merkle block sealing telemetry.
compliance	**Regulatory Compliance Radar**	Multi-jurisdictional compliance intelligence desk. Evaluates every supplier and part against US EAR99 export classifications, EU CBAM carbon border tariffs, and German LkSG human rights due diligence. Features instant legal citations and certificate verification.
operations	**Operations Conductor**	Tactical procurement and inventory execution desk. Displays real-time warehouse stock, active deficits, supplier bidding comparison grids, purchase order lifecycle progression, goods receipt notes (GRN), and dual-control approval queues.
alpha / whitepaper	**Competitive Alpha & Whitepaper**	Empirical benchmark evidence comparing AETHERIS AEOS to legacy incumbents (SAP S/4HANA, Palantir Foundry). Contains rigorous mathematical SLA penalty arbitrage formulas, Byzantine consensus proofs, and downloadable executive whitepapers.

05. UI Components & Interactive Shell

The interface incorporates eighteen specialized React components engineered for responsiveness, high aesthetic density, and strict compliance with defense ergonomics:

- **Sidebar (Sidebar.tsx):** Collapsible navigational anchor supporting pinned view states, route badges, and real-time pending approval counters.
- **TopBar (TopBar.tsx):** Global status header displaying backend connection health (:3001), active organization, session role badge, theme switcher, and command palette hotkey.
- **LoginPage & AuthModal (LoginPage.tsx, AuthModal.tsx):** Enterprise authentication gates featuring one-click demo persona switcher, password validation, and token issuance.
- **UserProfileMenu (UserProfileMenu.tsx):** Account management popover indicating user UUID, organization identifier, active permissions, and session logout controls.
- **CommandPalette (CommandPalette.tsx):** Omnipresent modal triggered via Ctrl+K / ■K allowing rapid navigation, instant recovery triggering, and parts catalog search.
- **ApiKeyModal (ApiKeyModal.tsx):** Client-side Google Gemini API key configuration modal, storing keys safely in browser local storage without server leakage.
- **MasterPromptModal (MasterPromptModal.tsx):** LLM behavioral inspection tool providing full visibility into foundational agent prompts, temperature, and policy instructions.
- **AuditProofModal (AuditProofModal.tsx):** Cryptographic proof inspector rendering visual Merkle tree inclusion paths, sibling hashes, and Ed25519 digital signatures.
- **ConsensusVotingMatrix (ConsensusVotingMatrix.tsx):** Real-time table displaying individual agent votes, confidence weights, mathematical rationale, and quorum thresholds.
- **Dual Theme Engine:** Supports *Linear Dark Obsidian* (default) and *Apple Light Studio* via dynamic CSS variable injection and local storage persistence.

06. Multi-Agent Swarm Orchestration Engine

Implemented in server_py/services/orchestrator.py, the autonomous multi-agent swarm operates across three deterministic tiers to detect, resolve, and seal enterprise supply chain disruptions:

Tier	Agent Name	Agent ID	Deterministic Core Responsibility
L0	**Apex Orchestrator**	apex-orchestrator	Mission decomposition, task graph generation, shortage identification, inter-agent synchronization, and BFT consensus synthesis.
L1	**Operations Conductor**	operations-conductor	Inventory deficit calculation (Required - OnHand), supplier bid evaluation across lead times and unit costs, PO draft creation.
L1	**Regulatory Sentinel**	regulatory-sentinel	Statutory audit against US EAR99, EU CBAM carbon border tariffs, and German LkSG human rights due diligence with legal citations.
L1	**Market Alpha Strategist**	market-alpha-strategist	Financial arbitrage modeling. Compares daily late-delivery SLA penalties against expedited freight premiums to optimize net profitability.
L2	**Cryptographic Auditor**	cryptographic-auditor	State transformation canonicalization (RFC 8785), SHA-256 event chaining, binary Merkle tree root calculation, and Ed25519 block signing.

Swarm Consensus & Fault Tolerance Rules

The swarm enforces a strict **67% weighted BFT supermajority quorum**. Each agent casts a signed vote (APPROVE, CONDITIONAL, REJECT) accompanied by a confidence factor (0.00 to 1.00) and mathematical rationale. If Regulatory Sentinel detects an embargo violation (EAR99 breach), its rejection carries veto power regardless of cost arbitrage.

07. Cryptographic Engine & Immutable Ledger

Implemented in server_py/crypto_engine.py, AEOS provides mathematical proof of system integrity without requiring public blockchain infrastructure. The subsystem utilizes four cryptographic foundations:

- **1. RFC 8785 Canonical JSON Serialization:** Standardizes payload serialization by enforcing alphabetical key ordering, zero arbitrary whitespace, and UTF-8 encoding. Eliminates serialization non-determinism across platforms.
- **2. SHA-256 Cryptographic Hash Chaining:** Every audit event records the SHA-256 hash of its predecessor. Any silent tampering or record deletion by database administrators immediately invalidates the entire downward chain.
- **3. Binary Merkle Tree Construction:** Events are organized into binary Merkle trees. If an odd number of leaf nodes exists at any tree level, the final node is duplicated to guarantee balanced binary tree computation.
- **4. Ed25519 Asymmetric Digital Signatures:** Sealed blocks are signed with an Ed25519 private key. Auditors can independently verify the Merkle root and all child inclusion proofs using only the embedded public key.

ZERO-KNOWLEDGE AUDIT VERIFICATION

Proof of Inclusion Verification: Clients verify that event *E* was sealed in Block *B* by providing a minimal sibling hash path: Path = [(Sibling_1, 'right'), (Sibling_2, 'left'), ...]. Traversal reconstructs the root hash in O(log N) time, verifying exact cryptographic presence.

08. Relational Database Schema (14 Models)

Governed by SQLAlchemy ORM in server_py/database.py, the schema consists of fourteen interconnected tables with enforced foreign key cascades, unique constraints, and audit immutability:

Model / Table	Primary Key	Key Relational Columns & Data Types	System Role & Domain Constraints
organizations	id (String)	name, code, created_at	Multi-tenant organization boundary.
users	id (String)	email, hashed_password, role, org_id	Authenticated system actors with RBAC roles.
products	id (String)	sku, name, unit, specs, eccn	Item master with export control classification.
inventory	id (String)	product_id, warehouse_code, on_hand, reserved	Real-time physical stock across facilities.
orders	id (String)	customer_name, product_id, quantity, status	Customer fulfillment demand records.
suppliers	id (String)	name, country, lead_time_days, rating	Global vendor master with geographic metadata.
purchase_orders	id (String)	po_number, supplier_id, quantity, total_amount	Procurement liabilities with line items.
agent_runs	id (String)	run_number, status, order_id, summary	Top-level execution pipeline runs.
agent_tasks	id (String)	run_id, agent_name, status, payload	Discrete agent tasks with input/output states.
approval_requests	id (String)	run_id, po_id, status, threshold_amount	Human-in-the-loop governance holding records.
compliance_docs	id (String)	supplier_id, doc_type, issuing_authority	Certificates of origin, CBAM declarations.
compliance_evals	id (String)	run_id, framework, passed, citations	Statutory legal audit reports.
audit_events	id (String)	event_type, actor, payload_hash, prev_hash	SHA-256 chained immutable audit ledger.
merkle_blocks	id (String)	block_number, merkle_root, signature, events_count	Sealed cryptographic blocks with Ed25519.

09. Authoritative REST API Reference

All endpoints are served asynchronously by FastAPI on port 3001. The table below details authoritative contracts:

Method & Endpoint	Role Required	Payload / Params	Response & Functional Behavior
POST /api/auth/login	Public	{ email, password }	Verifies password against salted bcrypt hash. Returns JWT session token and role profile.
POST /api/auth/switch-role	Authenticated	{ role }	Allows authorized tenant switching between demo personas for audit testing.
POST /api/orchestration/runs/start-recovery	Authenticated	{ order_id, notes }	Triggers 5-agent swarm disruption recovery. Spawns tasks, runs arbitrage, checks compliance.
GET /api/orchestration/runs/{id}	Authenticated	Path: id	Fetches granular run telemetry, agent logs, decision matrices, and task lifecycle statuses.
POST /api/orchestration/approvals/{id}/decide	Director / Exec	{ decision, reason }	Approves/rejects PO exceeding $5,000. Rejects analysts and self-approvers with HTTP 403.
POST /api/orchestration/receipts/confirm	Operations Director	{ po_id, received_qty }	Creates Goods Receipt Note (GRN), updates warehouse inventory (+200), triggers Merkle seal.
GET /api/orchestration/merkle/verify/{block}	Public	Path: block_number	Returns Merkle root, leaf nodes, sibling inclusion paths, and Ed25519 signature validity.
GET /api/health	Public	None	Returns system status, active database dialect, agent readiness, and version string.

10. Enterprise Security, RBAC & Governance

AETHERIS AEOS implements zero-trust authorization. Role capabilities and governance boundaries are strictly enforced:

Role Identifier	Title & Persona	View Radar & Cockpits	Trigger Recovery	Approve PO (>$5k)	Confirm GRN	Export Ledger
executive	Elena Rostova (COO)	Full Access	Yes	**YES**	Yes	**Full Export**
operations_director	Marcus Vance (Dir Ops)	Full Access	Yes	**YES**	**YES**	Full Export
compliance_officer	Sophia Chen (Counsel)	Full Access	Read Only	**Denied (403)**	**Denied (403)**	**Full Export**
analyst	Jordan Taylor (Analyst)	Full Access	Yes	**Denied (403)**	**Denied (403)**	View Only

DUAL CONTROL ENFORCEMENT

Separation of Duties (Four-Eyes Principle): If an operations director initiates a procurement recovery run, the backend cryptographically prevents self-approval. Dual control requires a distinct authorized user to execute the approval payload.

11. Statutory Regulatory Compliance

Every supplier bid and component procurement is audited against three major global statutory mandates:

- **US Export Administration Regulations (EAR99 - 15 CFR § 734):** Aerospace components (such as Titanium Grade 5 Flanges) are audited against the Commerce Control List (CCL) and OFAC Specially Designated Nationals lists to prevent unauthorized re-export or dual-use diversion.
- **EU Carbon Border Adjustment Mechanism (EU CBAM - Regulation 2023/956):** Audits embedded Scope 1 and Scope 2 carbon emissions per metric ton of imported titanium and iron under CN Code 8108.90. Calculates prospective CBAM certificate liabilities prior to purchase commitment.
- **German Supply Chain Due Diligence Act (LkSG - § 3 LkSG):** Mandates continuous human rights and environmental vetting across Tier-1 and direct Tier-2 suppliers. Validates valid ISO 14001:2015 environmental systems and verifies absence of forced labor infractions.

12. Benchmark Operational Walkthrough

The canonical operational benchmark demonstrates the real-time fulfillment of Order **ORD-2026-500** (Airbus Defense & Space requirement for 500 Titanium Flanges):

Step 1 - Shortage Ingestion: Airbus places ORD-2026-500 (500 units TF-700). Inventory holds 300 units in Central Warehouse. Apex Orchestrator detects acute 200-unit deficit.

Step 2 - Market Arbitrage: Operations Conductor queries supplier master. Selects *Bavaria Precision Metallics* (200 units @ $54.00/unit = $10,800 total; 2-day lead time) over Swedish and Chinese alternatives.

Step 3 - Statutory Vetting: Regulatory Sentinel verifies EAR99 export classification (PASSED), EU CBAM embedded emissions (1.42t CO2 PASSED), and German LkSG human rights certification (PASSED).

Step 4 - Governance Hold: Because PO total ($10,800) exceeds the autonomous $5,000 threshold, execution halts into an escrow approval hold (PENDING_APPROVAL).

Step 5 - RBAC Approval: Marcus Vance (Operations Director) executes cryptographic approval. An attempted approval by Jordan Taylor (Analyst) is rejected with HTTP 403.

Step 6 - Goods Receipt (GRN): Upon delivery, Marcus confirms receipt. Physical inventory increments to 500 units (+200). Order ORD-2026-500 is fully allocated and dispatched.

Step 7 - Merkle Sealing: Cryptographic Auditor bundles all 7 state mutations into canonical JSON, constructs a balanced binary Merkle tree, and seals Block #104 with an Ed25519 signature.

13. Mathematical Formulations & Resilience

Agent decision-making is strictly governed by deterministic mathematical formulations:

```
1. Net Arbitrage Utility Function: U(s) = [ (T_sla - T_lead(s)) * Penalty_daily ] - [ Q * P_unit(s) + C_freight(s) + C_cbam(s) ] Where: T_sla = 3 days | T_lead = 2 days | Penalty_daily = $2,500/day | Q = 200 units | P_unit = $54.00 Net Utility = (1 day * $2,500) - ($10,800 + $600 + $150) = +$13,450 relative to SLA default. 2. Byzantine Consensus Quorum Condition: Q_bft = Sum(w_i * v_i) / Sum(w_i) >= 0.67 (67% Supermajority requirement)
```

Adversarial Attack Resilience Benchmarks

Attack Scenario	Injected Threat Vector	Detection Mechanism	System Defense Response
SQL Record Mutation	Malicious DB admin alters unit price from $54 to $40 directly in database.	SHA-256 canonical hash chain re-computation.	Chain breaks at block #104. Security alarm raised; transaction rejected.
Role Escalation Attempt	User with Analyst token POSTs to /api/orchestration/approvals/{id}/decide.	FastAPI dependency authorization gate.	Immediate HTTP 403 Forbidden. Violation logged into immutable audit chain.
Merkle Tree Forgery	Attacker supplies forged sibling hash to claim false inclusion.	Binary Merkle root path traversal.	Calculated root does not match signed Ed25519 root. Proof rejected.

14. Developer Handbook & Deployment

To initialize and run AETHERIS AEOS in local or production development mode, execute the following commands:

```
# 1. Start FastAPI Authoritative Backend (Port 3001) cd "d:\DEVLOPEMENT\2nd project\2nd project" python -m uvicorn server_py.main:app --host 0.0.0.0 --port 3001 --reload # 2. Start Reactive Vite Frontend (Port 5173) cd "d:\DEVLOPEMENT\2nd project\2nd project" npm run dev
```

Default Enterprise Seed Credentials

Role Title	Email Identifier	Default Password	Governance Scope
Executive (COO)	elena@aetheris.ai	Executive2026!	Full oversight, unlimited approval ceiling, audit export
Operations Director	marcus@aetheris.ai	Operations2026!	Procurement lead, PO approval, Goods Receipt (GRN)
Compliance Officer	sophia@aetheris.ai	Compliance2026!	Regulatory radar lead, legal citation auditor
Supply Chain Analyst	jordan@aetheris.ai	Analyst2026!	Shortage initiator, operational reviewer (Read/Propose)